



OPEN Band gaps of elastic waves in 1-D dielectric phononic crystal with the flexoelectric and strain gradient effects consideration

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The propagation of Bloch waves in one dimensional phononic crystal consisting of dielectric elastic solids is studied with consideration of the strain gradient, inertial gradient and flexoelectric effects. The transfer matrixes for single layer and one-cell of phononic structure are derived based on the constitutive equations and the governing equations of dielectric elastic solids. The dispersion equation for Bloch waves is obtained by the application of periodic conditions for the generalized displacements and tractions considered within strain gradient theory of electro-elasticity. Based on the numerical solution of the derived dispersion equation, the influences of the micro-stiffness length scale, micro-inertial length scale and flexoelectric coefficients on the dispersion and the bandgap are discussed.

Keywords Flexoelectricity, Strain gradients, Inertial gradients, Phononic crystal, Bloch waves

An artificial composite material designed to control the propagation of elastic waves is called phononic crystal (PC). The so called one-dimensional PC is the periodic laminated structure. PC can be used to insulate vibration, reduce noise and filter acoustic waves because the elastic waves can pass through laminated structures by frequency-selective. In other words, the band gaps will appear when the elastic waves cannot pass through laminated structures in a certain frequency range. Therefore, researchers are very interested in studying the special properties of PC in the past decades. Sun and Wei¹ and Zhan and Wei² studied the influence of the interface conditions and anisotropy on band gap of PC by use of classical elastic theory. Li et al.³ presented a dynamically tunable mechanism of wave transmission in one-dimensional helicoidal PC. Liu et al.⁴ proposed hyperelastic transformation materials in the design of PC in order to achieve stable elastic band-gaps that do not vary with deformation. Jia et al.⁵ proposed a topology optimization methodology to maximize the band gap of acoustic waves at a specified central frequency, and used the Kriging-based algorithm to solve the complicated optimization problem. Méndez et al.⁶ determined the dependence of the effective elastic parameters as a function of the frequency by using a theory of homogenization. In this dynamic case of homogenization, it was found that the effective parameters can reproduce the exact dispersion relations for the acoustic modes that propagate along the periodicity direction of the crystal.

It is well-known the microstructure effects and the size effects cannot be ignored any more, for example, the Bloch waves in the micro- and nano-phononic crystal, or the wavelength of Bloch waves is comparable to the characteristic length of microstructure^{7,8}. In order to take the microstructure effects and the size effects into consideration, the generalized elastic theory with consideration of microstructure effects and the size effects should be used instead. These generalized elastic theories includes the couple stress theory^{9,10}, the strain gradient elastic theory^{11,12}, the micromorphic theory¹³, the microstretch theory¹⁴, the micropolar theory¹⁵ and the nonlocal theory¹⁶. Therefore, in nanoscale periodic laminates, the band structures of the anti-plane transverse elastic wave propagating normally based on the nonlocal elastic continuum theory was investigated by Yan et al.¹⁷. The influences of the size-effect of each sub-layer, the material and structural parameters on the cut-off frequency and the dense band zones were discussed in this paper. Zheng et al.¹⁸ researched the band structures of nanoscale PCs by using a meshfree local radial basis function collocation method (LRBFCM) and the validity of the LRBFCM had been proved. Li et al.^{19,20} studied the effects of microstructure parameters on the band gaps by using strain gradient elasticity theory and couple stress elasticity theory, respectively. Guo and Wei²¹ studied the propagation properties of in-plane elastic waves in a nano-scale N-type piezoelectric

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semiconductor material/piezoelectric dielectric material layered periodic composite. Both the mathematical model and the physical mechanisms were fundamental to the analysis and optimization of nano-scale energy harvesters. The researchers found that nonlocal effects soften the stiffness of nano-sized structures. Hosseini et al.²² proposed an analytical method for the analysis of the effects of the strain gradients on the frequency band structures and band-gaps of the elastic waves propagating in one-dimensional (1D) nano-sized phononic crystals (PCs). The effects of the small-scale parameters in the strain-gradient theory on the band structures and the band-gap characteristics were analyzed and discussed in detail. It was demonstrated that the strain gradients may significantly influence the frequency band structures and the band-gap characteristics of the elastic waves propagating in 1D nano-sized PCs.

Although the Bloch waves and their band gaps properties have been investigated in the past two decades, however, the studies on band gaps of elastic waves in 1-D dielectric phononic crystal with the flexoelectric and strain gradient effects consideration are rarely reported. The flexoelectric effect is the phenomenon of electrical polarization induced by strain gradient and result from the high-order mechanical and electrical coupling in dielectric elastic solid. The flexoelectric effect exists in almost all dielectric materials regardless of non-centrosymmetry^{23,24}. Usually, the flexoelectric effects is weak at macroscale but significantly be enhanced at micro and nano scale. When the wavelength of Bloch waves is comparable to the characteristic length of microstructure, the microstructure effects should not be ignored any more. At same time, the flexoelectric effects can also be ignored due to the evident increase of the strain gradient.

In this paper, the one-dimensional PC consisting of two different dielectric elastic solid which repeat periodically is studied with consideration of the flexoelectric effects. The transfer matrix of a typical single cell is derived on the basis of the constitutive equations and the perfect contact conditions at the interface of two different dielectric elastic solids. The dispersion equation is obtained by the application of the Bloch theorem and is solved numerically. The influence of the flexoelectric coefficient, the micro-stiffness length scale and the micro-inertial length scale parameter on the dispersion curves and the band gaps are discussed based on the numerical results.

Basic equations for the gradient theory in dielectrics

In the dielectric solid, constitutive equations with consideration of strain gradient and flexoelectric effects can be written as²⁵

$$\sigma_{ij} = c_{ijkl}\varepsilon_{kl}, \quad (1a)$$

$$\mu_{jkl} = -f_{ijkl}E_i + g_{jklmni}\eta_{nmi}, \quad (1b)$$

$$D_k = a_{kl}E_l + f_{klmn}\eta_{lmn}, \quad (1c)$$

where c_{ijkl} , f_{klmn} , g_{jklmni} and a_{kl} are the material property tensors. Symbols a_{kl} and c_{ijkl} represent the second-order and the fourth-order tensors of permittivity and elastic coefficients, respectively. f_{klmn} denotes the tensor of direct flexoelectric coefficients and two independent components f_1 and f_2 for f_{ijkl} are considered²⁶, namely, $f_{ijkl} = f_1\delta_{jk}\delta_{il} + f_2(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl})$. g_{jklmni} represents the tensor of the higher-order elastic coefficients, and g_{jklmni} are assumed to be proportional to the classical elastic stiffness coefficients c_{klmn} by the internal length material parameter l (micro-stiffness length scale parameter)^{27,28}, namely, $g_{jklmni} = l^2 c_{jklmn}\delta_{li}$. σ_{ij} , μ_{jkl} and D_k represent the stress, higher-order stress and electric displacement, respectively. ε_{ij} represent the strain tensor and are related to the displacements u_i , while E_j denote the electric field vector and are related to the electric potential ϕ , as follows

$$\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i}), \quad E_j = -\phi_{,j}. \quad (2)$$

The strain-gradient tensor η is defined as

$$\eta_{ijk} = \varepsilon_{ij,k} = \frac{1}{2}(u_{i,jk} + u_{j,ik}). \quad (3)$$

The most practically relevant dielectric materials are transversely isotropic, if the poling direction coincides with the material symmetry axis. In the case of a poling in the positive z -axis and x - y as isotropy plane the general problem may be separated into two 2D problems (plane and antiplane problem) and 1D problem. Let us consider 1D problem for $u_z(z, t)$ and $\phi(x_z, t)$ in the layered nanoscale phononic crystal structure exhibiting axial symmetry with respect to the third axis orthogonal to layers²⁹. Then,

$$\begin{aligned} \varepsilon_{xx} = \varepsilon_{yy} = \varepsilon_{xy} = \varepsilon_{xz} = \varepsilon_{yz} = 0, \quad \varepsilon_{zz} = u_{z,z}, \\ \eta_{xxk} = \eta_{yyk} = \eta_{xyk} = \eta_{xzk} = \eta_{yzk} = 0, \quad \eta_{zzz} = u_{z,zz}, \quad E_z = -\phi_{,z}. \end{aligned} \quad (4)$$

By using the standard Voigt notation for tensor material coefficients, the constitutive Eq. (1) can be written in matrix form as

$$\begin{pmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{yz} \\ \sigma_{xz} \\ \sigma_{xy} \end{pmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{13} & 0 & 0 & 0 \\ c_{13} & c_{13} & c_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.5(c_{11} - c_{12}) \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ u_{z,z} \\ 0 \\ 0 \\ 0 \end{pmatrix} = u_{z,z} \begin{pmatrix} c_{13} \\ c_{13} \\ c_{33} \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad (5)$$

$$\begin{pmatrix} \mu_{xxx} \\ \mu_{yyx} \\ \mu_{zzx} \\ \mu_{yxx} \\ \mu_{xzx} \\ \mu_{xyx} \end{pmatrix} = - \begin{bmatrix} f_1 + 2f_2 & 0 & 0 \\ f_1 & 0 & 0 \\ f_1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & f_2 \\ 0 & f_2 & 0 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ E_3 \end{pmatrix} + \mathbf{0} = - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ f_2 E_3 \\ 0 \end{bmatrix}, \quad (6a)$$

$$\begin{pmatrix} \mu_{xxy} \\ \mu_{yyy} \\ \mu_{zzy} \\ \mu_{yyz} \\ \mu_{xzy} \\ \mu_{xyy} \end{pmatrix} = - \begin{bmatrix} 0 & f_1 & 0 \\ 0 & f_1 + 2f_2 & 0 \\ 0 & f_1 & 0 \\ 0 & 0 & f_2 \\ 0 & 0 & 0 \\ f_2 & 0 & 0 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ E_3 \end{pmatrix} + \mathbf{0} = - \begin{bmatrix} 0 \\ 0 \\ 0 \\ f_2 E_3 \\ 0 \\ 0 \end{bmatrix}, \quad (6b)$$

$$\begin{pmatrix} \mu_{xxz} \\ \mu_{yyz} \\ \mu_{zzz} \\ \mu_{yzz} \\ \mu_{xzz} \\ \mu_{xyz} \end{pmatrix} = - \begin{bmatrix} 0 & 0 & f_1 \\ 0 & 0 & f_1 \\ 0 & 0 & f_1 + 2f_2 \\ 0 & f_2 & 0 \\ f_2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ E_3 \end{pmatrix} + l^2 \begin{bmatrix} c_{11} & c_{12} & c_{13} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{13} & 0 & 0 & 0 \\ c_{13} & c_{13} & c_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.5(c_{11} - c_{12}) \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ u_{z,zz} \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ = -E_3 \begin{bmatrix} f_1 \\ f_1 \\ f_1 + 2f_2 \\ 0 \\ 0 \\ 0 \end{bmatrix} + l^2 u_{z,zz} \begin{bmatrix} c_{13} \\ c_{13} \\ c_{33} \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad (6c)$$

$$\begin{pmatrix} D_x \\ D_y \\ D_z \end{pmatrix} = \begin{bmatrix} a_1 & 0 & 0 \\ 0 & a_1 & 0 \\ 0 & 0 & a_3 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ E_3 \end{pmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 & f_2 & 0 \\ 0 & 0 & 0 & f_2 & 0 & 0 \\ f_1 & f_1 & f_1 + 2f_2 & 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ u_{z,zz} \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ = \begin{bmatrix} 0 \\ 0 \\ a_3 E_3 + (f_1 + 2f_2) u_{z,zz} \end{bmatrix}. \quad (7)$$

Therefore, the constitutive equations can be simplified and rewritten as

$$\sigma_{zz} = c_{33} u_{z,z}, \quad (8)$$

$$\mu_{zzz} = l^2 c_{33} u_{z,zz} + (f_1 + 2f_2) \phi_{,z}, \quad (9)$$

$$D_z = -a_3 \phi_{,z} + (f_1 + 2f_2) u_{z,zz}. \quad (10)$$

The governing differential equations and boundary conditions for 1D electroelastic problem are derived from the Hamilton variational principle³⁰

$$\delta \int_{t_0}^{t_1} \int_V (K - U) dV dt + \int_{t_0}^{t_1} \int_S \delta W dS dt = 0, \quad (11)$$

where $\int_V K dV$ and $\int_V U dV$ are the total kinetic energy and internal energy, respectively, and $\int_S \delta W dS$ is the variation of the work done by external forces acting on the boundary S of volume V . Assuming low frequency mechanical loadings, the quasi-static approximation is allowed for the electric field.

The variation of internal energy per volume $\mathcal{U}$ in gradient theory is given by^{31,32}

$$\delta\mathcal{U} = \sigma_{ij}\delta\varepsilon_{ij} + \mu_{ijk}\delta\eta_{ijk} + D_k\delta\phi_{,k} = \sigma_{zz}\delta u_{z,z} + \mu_{zzz}\delta u_{z,zz} + D_z\delta\phi_{,z}. \quad (12)$$

Then,

$$\begin{aligned} \delta U &= \delta \int_V \mathcal{U}(\varepsilon_{ij}, \eta_{ijk}, \phi_{,k}) dV = \int_V (\sigma_{zz}\delta u_{z,z} + \mu_{zzz}\delta u_{z,zz} + D_z\delta\phi_{,z}) dV \\ &= \int_S [(\sigma_{zz} - \mu_{zzz,z})\delta u_z + n_z\mu_{zzz}\delta u_{z,z} + D_z\delta\phi] dS - \int_V [(\sigma_{zz,z} - \mu_{zzz,zz})\delta u_z + D_{z,z}\delta\phi] dV. \end{aligned} \quad (13)$$

The variation of kinetic energy contribution is described by Askes and Aifantis²⁷.

$$\begin{aligned} \delta \int_{t_0}^{t_1} \int_V K dV dt &= \int_{t_0}^{t_1} \int_V \rho (\dot{u}_z\delta\dot{u}_z + l_1^2\dot{u}_{z,z}\delta\dot{u}_{z,z}) dV dt = - \int_{t_0}^{t_1} \int_V \rho (\ddot{u}_z\delta u_z + l_1^2\ddot{u}_{z,z}\delta u_{z,z}) dV dt \\ &= \int_{t_0}^{t_1} \left[- \int_S \rho l_1^2 \ddot{u}_{z,z} \delta u_z dS - \int_V \rho (\ddot{u}_z - l_1^2 \ddot{u}_{z,zz}) \delta u_z dV \right] dt, \end{aligned} \quad (14)$$

where ρ is the mass density and l_1 is the micro-inertial length scale parameter.

The variation of work done by external forces can be written as

$$\delta W = \int_V F_k \delta u_k dV + \int_S P_k \delta u_k dS + \int_S R_k n_l \delta u_{k,l} dS, \quad (15)$$

where F_k is the body force and is ignored here. P_k is the monopolar traction, R_k is the dipolar tractions¹⁹.

Inserting Eqs. (13), (14) and (15) into Eq. (11) leads to the following equation.

$$\begin{aligned} &\delta \int_{t_0}^{t_1} \int_V (K - U) dV dt + \int_{t_0}^{t_1} \left(\int_S P_k \delta u_k dS + \int_S R_k n_l \delta u_{k,l} dS \right) dt \\ &= \int_{t_0}^{t_1} \left\{ \int_S \left\{ [P_z - (\sigma_{zz} - \mu_{zzz,z} + \rho l_1^2 \ddot{u}_{z,z})] \delta u_z + n_z (R_z - \mu_{zzz}) \delta u_{z,z} - D_z \delta\phi \right\} dS \right\} dt \\ &\quad + \int_{t_0}^{t_1} \left\{ \int_V [(\sigma_{zz,z} - \mu_{zzz,zz}) - \rho (\ddot{u}_z - l_1^2 \ddot{u}_{z,zz}) \delta u_z + D_{z,z} \delta\phi] dV \right\} dt. \end{aligned} \quad (16)$$

From Eq. (16), the governing equation and the boundary conditions can be obtained.

$$\sigma_{zz,z} - \mu_{zzz,zz} = \rho (\ddot{u}_z - l_1^2 \ddot{u}_{z,zz}), \quad (17)$$

$$D_{z,z} = 0, \quad (18)$$

$$P_z = \sigma_{zz} - \mu_{zzz,z} + \rho l_1^2 \ddot{u}_{z,z}, \quad (19)$$

$$R_z = \mu_{zzz}, \quad (20)$$

$$D_z = 0. \quad (21)$$

Substituting Eqs. (8)–(10) into Eqs. (17) and (18), the governing equations become

$$c_{33} (1 - l^2 \partial_z^2) \partial_z^2 u_z - f \partial_z^3 \phi = \rho (1 - l_1^2 \partial_z^2) \ddot{u}_z, \quad (22)$$

$$a_3 \partial_z^2 \phi - f \partial_z^3 u_z = 0, \quad (23)$$

where $f = (f_1 + 2f_2)$; $\partial_z^k(\cdot) = \frac{\partial^k}{\partial z^k}(\cdot)$.

In order to solve the system of Eqs. (22) and (23), it can be rewritten as

$$[a_3 c_{33} - (a_3 c_{33} l^2 + f^2) \partial_z^2] \partial_z^2 u_z - a_3 \rho (1 - l_1^2 \partial_z^2) \ddot{u}_z = 0, \quad (24)$$

$$\phi = A_1 + A_2 z + \frac{f}{a_3} \partial_z u_z. \quad (25)$$

with A_1 and A_2 being integration constants.

In contrast to the classical elasticity (when $l = l_1 = f = 0$), the higher order derivatives affect the governing Eq. (24).

The wave motion solution of Eq. (24) is

$$u_z = H \exp i(kz - \omega t). \quad (26)$$

Inserting Eq. (26) into Eq. (24), yields

$$(a_3 c_{33} l^2 + f^2) k^4 + (a_3 c_{33} - a_3 \rho \omega^2 l_1^2) k^2 - a_3 \rho \omega^2 = 0, \quad (27)$$

the roots of Eq. (27)

$$k_1 = -k_3 = \left[\frac{(a_3 \rho \omega^2 l_1^2 - a_3 c_{33}) + \sqrt{(a_3 c_{33} - a_3 \rho \omega^2 l_1^2)^2 + 4(a_3 c_{33} l^2 + f^2) a_3 \rho \omega^2}}{2(a_3 c_{33} l^2 + f^2)} \right]^{1/2}, \quad (28)$$

$$k_2 = -k_4 = \left[\frac{(a_3 \rho \omega^2 l_1^2 - a_3 c_{33}) - \sqrt{(a_3 c_{33} - a_3 \rho \omega^2 l_1^2)^2 + 4(a_3 c_{33} l^2 + f^2) a_3 \rho \omega^2}}{2(a_3 c_{33} l^2 + f^2)} \right]^{1/2}, \quad (29)$$

provides the dispersion relations $k_\alpha = k_\alpha(\omega)$, ($\alpha = 1, 2, 3, 4$). Omitting the flexoelectric effect, the dispersion relations become

$$k_1 = -k_3 = \left[\frac{(\rho \omega^2 l_1^2 - c_{33}) + \sqrt{(c_{33} - \rho \omega^2 l_1^2)^2 + 4c_{33} l^2 \rho \omega^2}}{2c_{33} l^2} \right]^{1/2}, \quad (30)$$

$$k_2 = -k_4 = \left[\frac{(\rho \omega^2 l_1^2 - c_{33}) - \sqrt{(c_{33} - \rho \omega^2 l_1^2)^2 + 4c_{33} l^2 \rho \omega^2}}{2c_{33} l^2} \right]^{1/2}. \quad (31)$$

Equations (30) and (31) mean that there are four coupled elastic waves which are classified as two groups with wave numbers $\pm k_1$ and $\pm k_2$. Two waves in a group have the same speed but opposite propagation direction. Thus, the general solution of Eq. (24) is given by superposition

$$u_z(z, t) = \sum_{\alpha=1}^4 H_\alpha e^{i(k_\alpha z - \omega t)}, \quad (32)$$

where H_α , ($\alpha = 1, 2, 3, 4$) are the displacement amplitudes.

According to (25) and (32), the general solution for electric potential is given as

$$\phi(z, t) = A_1 + A_2 z + \frac{f}{a_3} u_{z,z}(z, t), \quad (33)$$

where $u_{z,z}(z, t) = ik_\alpha \sum_{\alpha=1}^4 H_\alpha e^{i(k_\alpha z - \omega t)}$.

In Eq. (25), the integration constants A_2 becomes zero from the periodicity requirement, and A_1 can be taken arbitrary (say $A_1 = 0$, because constant electric potential yields zero electric field). Thus,

$$\phi(z, t) = \frac{f}{a_3} u_{z,z}(z, t). \quad (34)$$

Furthermore, in view of Eqs. (8)–(10), we obtain the solution for the rest of field variables within the interval $z \in [a, b]$

$$\sigma_{zz}(z, t) = c_{33} u_{z,z}(z, t), \quad (35)$$

$$\mu_{zzz}(z, t) = \frac{1}{a_3} (l^2 c_{33} a_3 + f^2) u_{z,zz}(z, t), \quad (36)$$

$$D_z(z, t) \equiv 0, \quad (37)$$

where $u_{z,zz}(z, t) = -k_\alpha^2 \sum_{\alpha=1}^4 H_\alpha e^{i(k_\alpha z - \omega t)}$.

Bearing in mind (34) and (37), the set of independent boundary quantities $\{u_z, u_{z,z}, \phi, P_z, R_z, D_z\}$ at the end points of the interval $[a, b]$ is reduced to $\{u_z, u_{z,z}, P_z, R_z\}$ since $\phi = (f/a_3) u_{z,z}$ and $D_z \equiv 0$. Furthermore, making use the general solution, we can write

$$u_z|_{z=a,b} = \sum_{\alpha=1}^4 H_\alpha e^{i(k_\alpha z - \omega t)} \Big|_{z=a,b}, \quad (38)$$

$$u_{z,z}|_{z=a,b} = \sum_{\alpha=1}^4 ik_\alpha H_\alpha e^{i(k_\alpha z - \omega t)} \Big|_{z=a,b}, \quad (39)$$

$$R_z|_{z=a,b} = \mu_{zz}|_{z=a,b} = -\sum_{\alpha=1}^4 k_{\alpha}^2 \left(l^2 c_{33} + \frac{f^2}{a_3} \right) H_{\alpha} e^{i(k_{\alpha} z - \omega t)} \Big|_{z=a,b}, \quad (40)$$

$$\begin{aligned} P_z|_{z=a,b} &= \sigma_{zz}|_{z=a,b} - \mu_{zz,z}|_{z=a,b} - l_1^2 \omega^2 u_{z,z}|_{z=a,b} \\ &= \sum_{\alpha=1}^4 i k_{\alpha} \left[c_{33} - l_1^2 \omega^2 + k_{\alpha}^2 \left(l^2 c_{33} + \frac{f^2}{a_3} \right) \right] H_{\alpha} e^{i(k_{\alpha} z - \omega t)} \Big|_{z=a,b}, \end{aligned} \quad (41)$$

Dispersion relation of a Bloch wave in 1-D phononic structures

Consider a periodic binary laminated structure, namely, one-dimension phononic crystal. The thickness of layer A is a_A and the thickness of layer B is a_B . The thickness of one single cell is $a_A + a_B$, see Fig. 1.

The displacement field in any layer of a single cell is formed by the forward and the backward bulk waves propagating along the normal of interfaces. Define the state vector of independent boundary quantities as

$$\{V\} = \{ u_z \quad u_{z,z} \quad P_z \quad R_z \}^T. \quad (42)$$

Let z_j^L be the axial coordinate of the left side of the layer “j” ($j = A$ or B) and z_j^R be the right side. Then, the axial coordinate of the right side of the layer is $z_j^R = z_j^L + a_j$, according to Eq. (42), the state vector at the left and right boundaries of a layer can be expressed as

$$\{V(z_j^L)\} = [P(z_j^L)] \{ H_1 \quad H_2 \quad H_3 \quad H_4 \}^T e^{-i\omega t}, \quad (43)$$

$$\{V(z_j^R)\} = [P(z_j^R)] \{ H_1 \quad H_2 \quad H_3 \quad H_4 \}^T e^{-i\omega t}, \quad (44)$$

the relation of matrixes $[P(z_j^L)]$ and $[P(z_j^R)]$ is

$$[P_j^R(z)] = [P_0] \cdot [G(z_j^R)] = [P_0] \cdot [G(z_j^L)] [G(a_j)] = [P_j^L(z)] \cdot [G(a_j)], \quad (45)$$

where

$$\begin{aligned} [P_0] &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ ik_1 & ik_2 & ik_3 & ik_4 \\ d_1 & d_2 & d_3 & d_4 \\ -k_1^2 \left(l^2 c_{33} + \frac{f^2}{a_3} \right) & -k_2^2 \left(l^2 c_{33} + \frac{f^2}{a_3} \right) & -k_3^2 \left(l^2 c_{33} + \frac{f^2}{a_3} \right) & -k_4^2 \left(l^2 c_{33} + \frac{f^2}{a_3} \right) \end{bmatrix} \\ d_s &= ik_s \left[c_{33} - l_1^2 \omega^2 + k_s^2 \left(l^2 c_{33} + \frac{f^2}{a_3} \right) \right] \quad (s = 1, 2, 3, 4), \end{aligned}$$

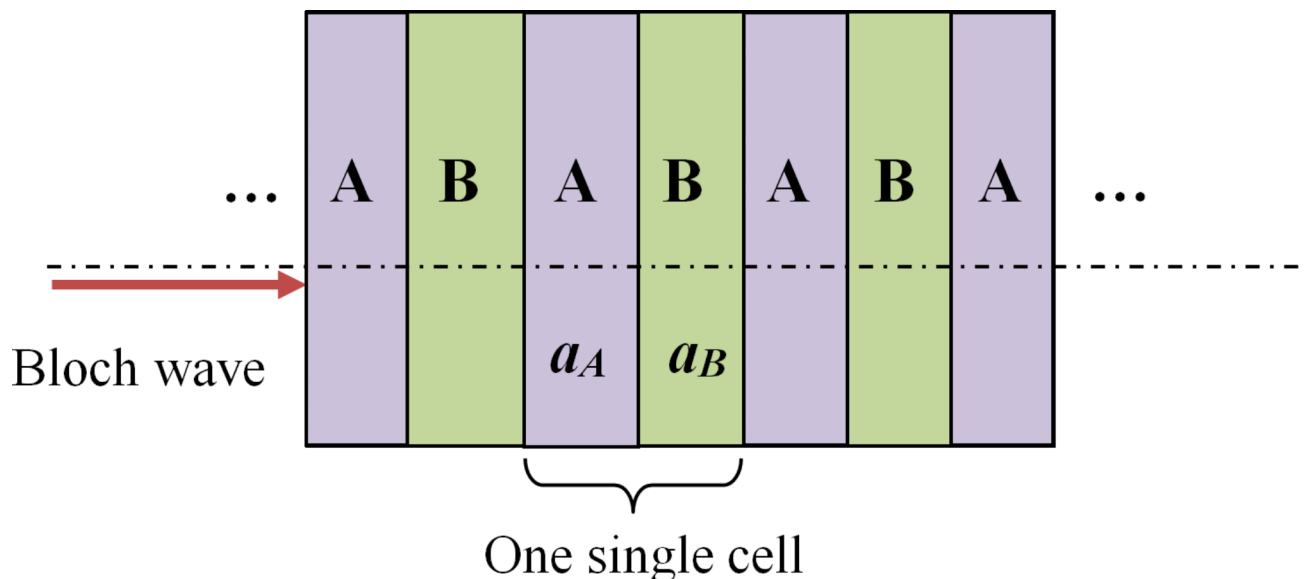


Fig. 1. The sketch of a typical single cell in the periodic laminated structure of dielectric elastic solid.

$$[G(z)] = \begin{bmatrix} e^{ik_1 z} & 0 & 0 & 0 \\ 0 & e^{ik_2 z} & 0 & 0 \\ 0 & 0 & e^{ik_3 z} & 0 \\ 0 & 0 & 0 & e^{ik_4 z} \end{bmatrix}.$$

One can easily find the correlation between the state vectors at the left and right sides of the j -layer

$$\{V(z_j^R)\} = [P(z_j^R)] \{H_1 \ H_2 \ H_3 \ H_4\}^T e^{-i\omega t} = [P(z_j^R)] [P(z_j^L)]^{-1} \{V(z_j^L)\} = [T_j] \{V(z_j^L)\}, \quad (46)$$

According to Eqs. (45) and (46), the following equation is obtained

$$[P(z_j^R)] [P(z_j^L)]^{-1} = [P_0] [G(z_j^R)] [G(z_j^L)]^{-1} [P_0]^{-1}, \quad (47)$$

Therefore, the transfer matrix can be defined by

$$[T_j] = [P_0] [G(a_j)] [P_0]^{-1}. \quad (48)$$

The transfer matrix $[T_j]$ is determined by the material coefficients, the thickness of the layer and the wave modes in the j -layer for a given frequency ω .

For the perfect interface contact, the state vector is continuous across the interface between layer A and layer B, namely,

$$\{V(z_B^L)\} = \{V(z_A^R)\}. \quad (49)$$

This interface condition leads to

$$\{V(z_B^R)\} = [T_B] [T_A] \{V(z_A^L)\} \quad (50)$$

The Bloch theorem for the wave propagation in the periodic structure can be expressed as

$$\{V(z_B^R)\} = e^{ikb} \{V(z_A^L)\}, \quad (51)$$

where $b = a_A + a_B$ is the thickness of a typical single cell, and k is the wavenumber of Bloch wave in the periodic laminated structure.

Inserting Eq. (50) into Eq. (51) leads to

$$([T_B] [T_A] - e^{ikb} [I]) \{V(z_A^L)\} = \{0\}, \quad (52)$$

The existence of non-trivial solution requires

$$\det ([T_B] [T_A] - e^{ikb} [I]) = f(\omega, k) = 0. \quad (53)$$

Equation (53) is the dispersion relation of Bloch wave.

Numerical results and discussions

The dispersion relation (the dependence of wavenumber k on the angular frequency ω) of Bloch waves in the periodic laminated structure is dependent on: (1) the thickness (a_A, a_B) of two dielectric elastic solids; (2) the material constants ($\rho, a_{33}, c_{33}, f, l, l_1$) of two dielectric elastic solids. In general, the dispersion equation can be written as

$$f(\rho, a_3, c_{33}, f, l, l_1, a_A, \rho', a'_3, c'_{33}, f', l', l'_1, a_B, k, \omega) = 0. \quad (54)$$

where ($\rho, a_3, c_{33}, f, l, l_1$) and a_A are the material coefficients in layer A, while ($\rho', a'_3, c'_{33}, f', l', l'_1$) and a_B are the material coefficients in layer B.

Choosing (ρ, a_3, b) and $\omega_0 = 2\pi / \left(\frac{a_A}{\sqrt{c_{33}/\rho}} + \frac{a_B}{\sqrt{c'_{33}/\rho'}} \right)$ as the reference values for material coefficients, we can define the non-dimensional material coefficients by

$$\bar{\rho} = \frac{\rho'}{\rho}, \bar{a}_3 = \frac{a'_3}{a_3}, \bar{c}_{33} = \frac{c'_{33}}{c_{33}}, \bar{f} = \frac{f}{f}, \bar{l} = \frac{l}{l}, \bar{l}_1 = \frac{l_1}{l_1}, \bar{a}_A = \frac{a_A}{a_A}, \bar{c} = \frac{c'_{33}}{c_{33}}, \bar{F} = \frac{f'}{f}, \bar{L} = \frac{l'}{l}, \bar{L}_1 = \frac{l'_1}{l_1}, \bar{a}_B = \frac{a_B}{a_A}, \bar{k} = \frac{kb}{\pi},$$

$$\bar{\omega} = \frac{\omega}{\omega_0}.$$

and the dispersion equation can be rewritten as

$$f(1, 1, \bar{c}_{33}, \bar{f}, \bar{l}, \bar{l}_1, \bar{a}_A, \bar{\rho}, \bar{a}_3, \bar{c}, \bar{F}, \bar{L}, \bar{L}_1, \bar{a}_B, \bar{k}, \bar{\omega}) = 0. \quad (55)$$

In this paper, ALN and BaTiO₃ are selected as the two solid materials to compose 1-D phononic crystals, namely the material of layer A is ALN and the material of layer B is BaTiO₃, and their physical constants are respectively³³, $\rho = 3.23 \times 10^3 \text{ kg/m}^3, a_3 = 8.4 \times 10^{-11} \text{ C}^2/\text{Nm}^2, c_{33} = 3.9 \times 10^{11} \text{ Pa}, a_A = a_B = 0.01 \text{ m}$,

$$\rho' = 5.8 \times 10^3 \text{ kg/m}^3, a_3' = 1.26 \times 10^{-8} \text{ C}^2/\text{Nm}^2, c_{33}' = 1.62 \times 10^{11} \text{ Pa}.$$

Figure 2 gives the comparison of dispersion curves of the one-dimension phononic crystal with and without the gradient effects and flexoelectric effects. It is observed that in the strain gradient elasticity, the dispersion curves shift toward high frequency region while the consideration of flexoelectricity results in the shift of the dispersion curves toward to low frequency region. When the strain gradients are considered, the bandgap at the middle and the margin of the Brillouin zone open evidently. When the flexoelectric effects are included, the dispersion curves become more flat at the Brillouin zone and the width of the bandgap at the middle of Brillouin zone is increased evidently. However, the bandgap at margin of Brillouin zone become narrow on the contrary due to the shifting toward low frequency of dispersion curves.

Figure 3 shows the influences of micro-stiffness length scale parameter l on the dispersion curves. It is observed that the dispersion curves shift toward high frequency region with increasing the strain gradient effects. This can be explained by enhancing the material stiffness with increasing the strain gradient effects. It is also observed that the dispersion curves at high frequency region are influenced more evidently than those at low frequency region. This phenomenon implies that strain gradient effects are more evident at high frequency region than at low frequency region. Therefore, gradient effects cannot be ignored for propagation of high frequency elastic waves.

Figure 4 shows the influence of the flexoelectric parameter f on the dispersion curves. It is observed that the dispersion curves shift gradually toward low frequency with increasing the flexoelectric parameter f . By comparison with the influence of the strain gradient effects, it is found that the flexoelectric effects are opposite. In fact, the flexoelectric effects enhance the polarization of dielectric material on one hand side and also make a contribution to the high order stress on the other hand side. Recall that the direct flexoelectricity is impossible without strain gradients and the polarization consumes certain portion of exerted energy. Bearing in mind the energy balance, the flexoelectric effect is partially competitive to strain gradient effect.

Figure 5 shows the influence of the micro-inertial length scale parameter l_1 on dispersion curves. This parameter plays important role in dynamic problems in small scale structures, especially for the wave propagation problems³⁴. If the inertial gradient parameter is ignored, the propagation speed of elastic waves could grow infinitely with the increasing frequency. The consideration of the micro-inertial effect can remedy the problem. It is observed that the dispersion curves shift toward the low frequency region and the dispersion curves become more flat with increasing the micro-inertial length scale parameter l_1 . By comparison of the Figs. 3 and 5, it is observed that the influences of the micro-inertial effect and the strain gradient effect are opposite.

Conclusions

The strain gradient effects and flexoelectric effects reflect the influences of microstructure of materials in phenomenological theory of continuous media. When the wavelength is close to the characteristic length of the microstructure, the influence of the microstructure on the propagation feature of elastic wave is not negligible. In the present work, the influence of microstructure parameters, such as the micro-stiffness length scale parameter, the micro-inertial length scale parameter and the flexoelectric coefficients, on the one-dimensional Bloch wave propagation in elastic dielectric crystal are investigated simultaneously. Based on the analytic expression of the dispersion equation and its numerical solution, the following conclusions can be drawn.

- 1) There are two independent direct flexoelectric coefficients f_1 and f_2 . However, from the constitutive equation, these two parameters f_1 and f_2 have the same influence on the one-dimensional phonon band gap, so the two parameters are treated as one parameter f in the case of one-dimensional problem.
- 2) The consideration of strain gradients results in enhancement of the dispersion feature of one-dimensional Bloch wave in phononic crystal. The dispersion curves shift toward high frequency region with increasing the strain gradient effects. By comparison with the classical elastic phononic crystal, the strain gradient effects help to open the bandgap.
- 3) The micro-inertial gradient effects on the dispersion of Bloch waves are opposite compared with the strain gradient effects. The strain gradients enhance the stiffness of material while the micro-inertial terms enhance the inertia of elastic movements in material. Both the micro-stiffness and micro-inertial effects are important for the wave propagation in small scale structures.
- 4) The direct flexoelectric effect results in the enhancement of the polarization due to the high order stresses in dielectric solid. The influences of polarization is opposite with that of micro-stiffness. Therefore, the flexoelectric effect and micro-stiffness effects result in a competitive relation on the dispersion of Bloch waves in layered dielectric solid.

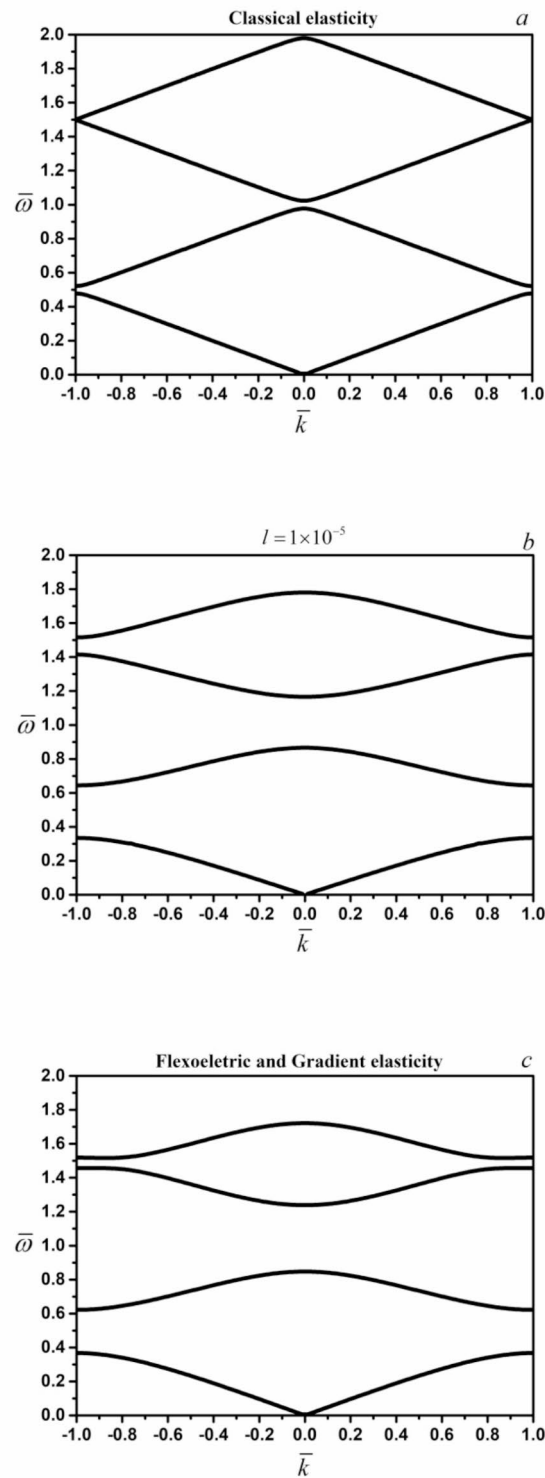


Fig. 2. Comparison of the dispersion curves and band gaps of dielectric phononic crystal based on three models. **a** Classical elasticity ($l = 0, l_1 = 0, f = 0, L = 0, L_1 = 0, F = 0$); **b** gradient elasticity ($l = 10^{-5}, l_1 = 2 \times 10^{-5}, f = 0, L = 5, L_1 = 5, F = 0$); **c** flexoelectric and gradient elasticity ($l = 10^{-5}, l_1 = 2 \times 10^{-5}, f = 10^{-5}, L = 5, L_1 = 5, F = 20$).

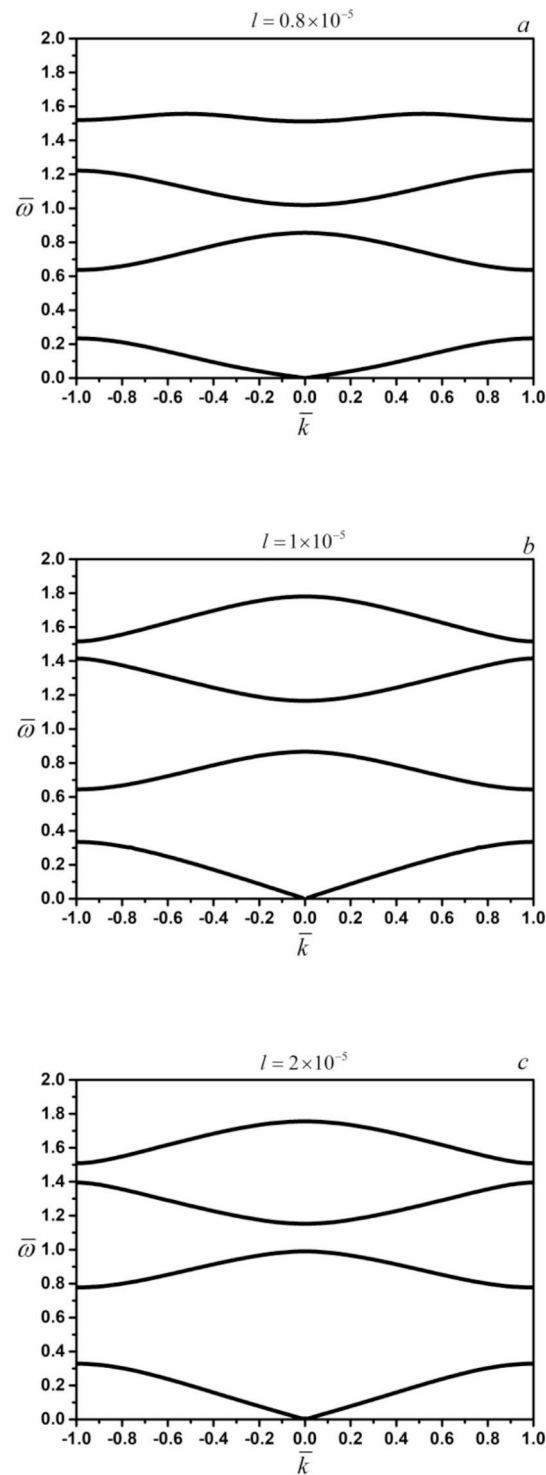


Fig. 3. The influence of micro-stiffness length scale parameter l on the dispersion curves and the band gaps in the flexoelectric gradient model ($l_1 = 1 \times 10^{-5}$, $f = 10^{-5}$, $L = 20$, $L_1 = 5$, $F = 200$).

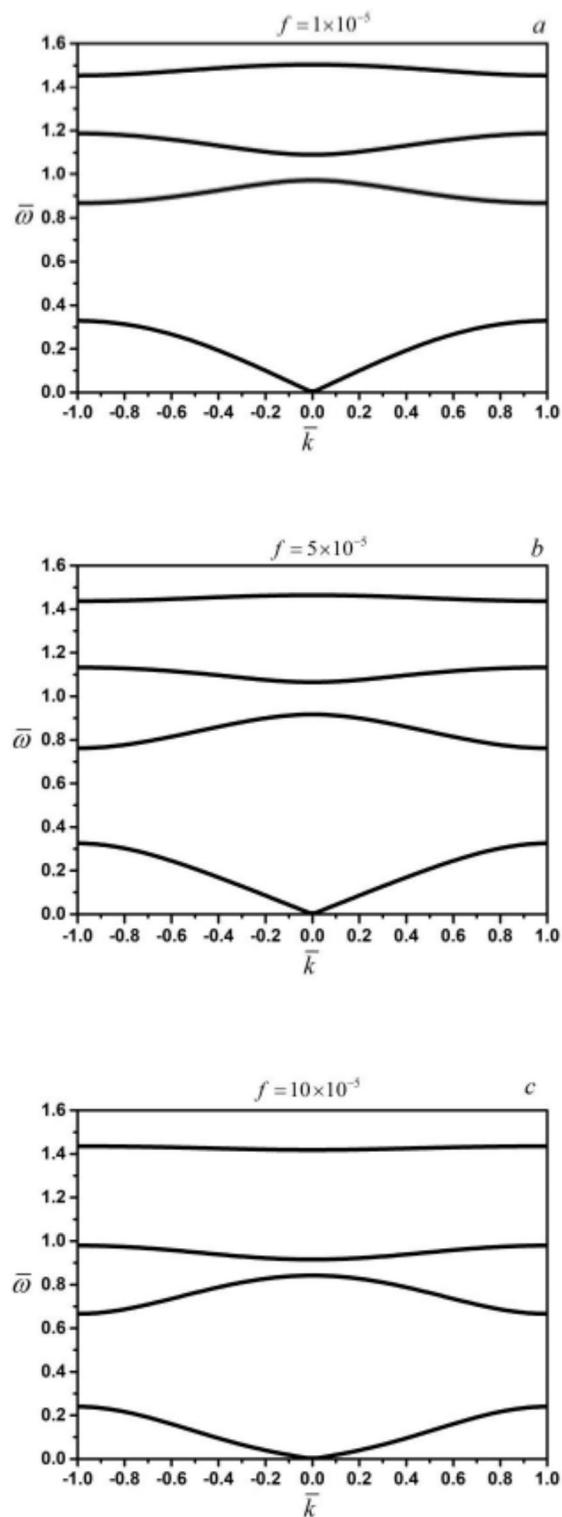


Fig. 4. The influence of flexoelectric coefficient f on the dispersion curves and the band gaps in the flexoelectric and gradient model ($l = 4 \times 10^{-5}$, $l_1 = 1 \times 10^{-5}$, $L = 20$, $L_1 = 5$, $F = 200$).

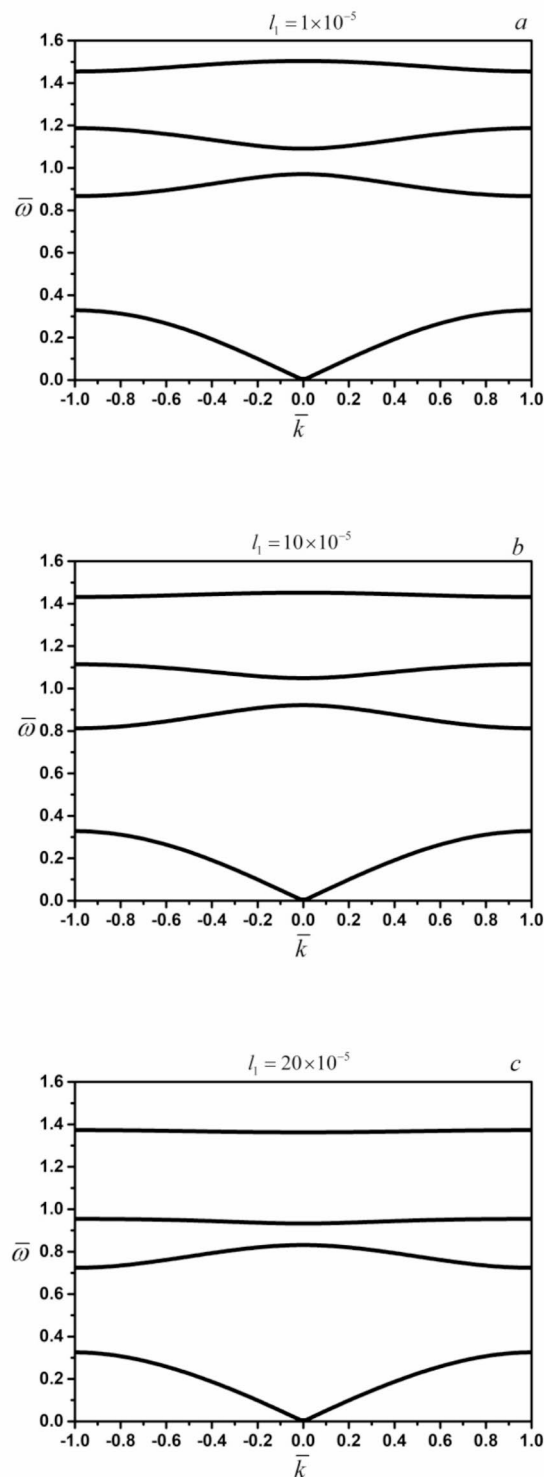


Fig. 5. The influence of the micro-inertial length scale parameter l_1 on the dispersion curves and the band gaps in the flexoelectric gradient model ($l = 4 \times 10^{-5}$, $l_1 = 1 \times 10^{-5}$, $f = 10^{-5}$, $L = 20$, $F = 200$, $L_1 = 5$).

Data availability

The datasets used and analysed during the current study can be obtained from the corresponding author. The email address of the corresponding author is liyueqiu2004@163.com.

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Author contributions

Ying Li and Yueqiu Li do the theoretic analysis and write this manuscript. Zihao Guo and Hong Wang do numerical discussion, Changda Wang writes the introduction and references of the manuscript. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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